QF632 - Financial Data Science

Building Credit Card Fraud Prediction and Security Risk Monitoring Model

Group 4

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Background Information

Why this topic matter?

Credit Card Profitability

Credit cards generate ~15% of issuer profits through fees and interest, making them a key revenue driver.

Market Growth and Trends

Consumer spending is expected to grow at 8.5% CAGR, boosted by e-commerce and payment tech innovation.

Fraud Risks in Retail

Small-value frauds (under $100) dominate, with projected $43B in losses—posing risks for everyday credit card users.

Customer Preservation

Banks must implement early fraud detection to avoid financial loss and preserve customer trust.

02

Data Extraction and Cleaning

Data Extraction

Data source → kaggle.com

	trans_date_trans_time	merchant	category	amt	city	state	lat	long	city_pop	job	dob	trans_num	merch_lat	merch_long	is_fraud
0	04-01-2019 00:58	"Stokes, Christiansen and Sipes"	grocery_net	14.37	Wales	AK	64.7556	-165.6723	145	"Administrator, education"	09-11-1939	a3806e984cec6ac0096d8184c64ad3a1	65.654142	-164.722603	1
1	04-01-2019 15:06	Predovic Inc	shopping_net	966.11	Wales	AK	64.7556	-165.6723	145	"Administrator, education"	09-11-1939	a59185fe1b9ccf21323f581d7477573f	65.468863	-165.473127	1
2	04-01-2019 22:37	Wisozk and Sons	misc_pos	49.61	Wales	AK	64.7556	-165.6723	145	"Administrator, education"	09-11-1939	86ba3a888b42cd3925881fa34177b4e0	65.347667	-165.914542	1
3	04-01-2019 23:06	Murray-Smitham	grocery_pos	295.26	Wales	AK	64.7556	-165.6723	145	"Administrator, education"	09-11-1939	3a068fe1d856f0ecedbed33e4b5f4496	64.445035	-166.080207	1
4	04-01-2019 23:59	Friesen Lt	health_fitness	18.17	Wales	AK	64.7556	-165.6723	145	"Administrator, education"	09-11-1939	891cdd1191028759dc20dc224347a0ff	65.447094	-165.446843	1

Some important features

ⓐ **Transactions time and date**
Provides insights into temporal patterns of fraudulent activity, help flag suspicious behavior

ⓑ **Merchant category**
Acts as risk profiling feature, such as certain merchants may become usual target by fraudsters

ⓒ **Transactions amount**
Helps detect atypical behavior, which often involve unusual large or small amounts

ⓓ **Credit card holder age**
Supports customer segmentation fraud targeting analysis to detect unusual pattern

ⓔ **Distance**
Calculates the distance between transactions and merchant to detect potential anomalies

ⓕ **Fraud labelling**
as target variable that we would like to predict using the other features.

Data Cleaning

Why does it matter?

Any inconsistencies, incomplete, or errors in data format might heavily influence the performance of the prediction models, leading to flawed results. This steps also involve data transformation into more suitable format for model training and analysis process.

Target variable

Variable : 'is_fraud'
Data type: binary label of [0, 1]

```
transaction_data['is_fraud'].unique()

array(['1', '1"2020-12-24 16:56:24"', '0', '0"2019-01-01 00:00:44"'],
      dtype=object)
```

Raw data has **several unique values** of fraud labelling with mixed of data types (i.e., string, numerical, etc)

Solution?

Convert into **string** type and extract only the first digit of each string using **regex expression.** Output values are 0 and 1, representing non-fraud and fraud transactions

Time and dates

Variable : 'trans_date_trans_time'
Data type: date time

trans_date_trans_time
04-01-2019 00:58

Need to separate the transaction time and date

Solution?

Use '**datetime' date, hour, dayofweek** to convert data into date, hour, and numerical data for days (0-6 for Mon-Sun)

```
df_cleaned['is_weekend'].unique()

array([0, 1])
```

Additional variables

Add **'is_weekend'** to indicate whether the transaction is taken place during weekdays/weekends

Missing timestamps

	trans_date_trans_time
24	NaT
25	NaT

is_fraud	date	hour	day_of_week	is_weekend
1	NaT	NaN	NaN	0
1	NaT	NaN	NaN	0

Some transactions have **no timestamp** information, results in **NaN values** in time-based features created

Solution ?

Create new binary column of **'missing_time'** to flag transactions that lack timestamp data.

Model could learn any absence itself may be predictive (i.e., **missingness as a feature**), also **avoid any skewed** predictions

Data Cleaning

Why does it matter?

Any inconsistencies, incomplete, or errors in data format might heavily influence the performance of the prediction models, leading to flawed results. This steps also involve data transformation into more suitable format for model training and analysis process.

Credit holder age

Variable : 'dob'
Data type: datetime

	dob	age
0	1939-09-11	86
1	1939-09-11	86
2	1939-09-11	86
3	1939-09-11	86
4	1939-09-11	86

Based on '**DOB**, we create new 'age' variable by subtracting with current year **(datetime.now().year)**

14441	NaT	60
14442	1956-01-09	69
14443	NaT	60
14444	1939-09-11	86
14445	NaT	60

Some has **no DOB** which lead to **NaN values** in age

Solution?
Use **average values** for neutrality and avoid data loss for input features

Distance info

Variable : 'lat, long, merch_lat, merch_long'
Data type: float

```
def haversine(lat1, lon1, lat2, lon2):
    R = 6371  # Earth radius in km
    lat1, lon1, lat2, lon2 = map(radians, [lat1, lon1, lat2, lon2])
    dlat = lat2 - lat1
    dlon = lon2 - lon1
    a = sin(dlat/2)**2 + cos(lat1) * cos(lat2) * sin(dlon/2)**2
    c = 2 * atan2(sqrt(a), sqrt(1-a))
    return R * c
```

Method?
Apply the **Haversine** formula which calculates the great circle distance using two geographic points (latitude and longitude)

Variables encoding

Variable : 'merchant, category, city, state, job'
Data type: categorical

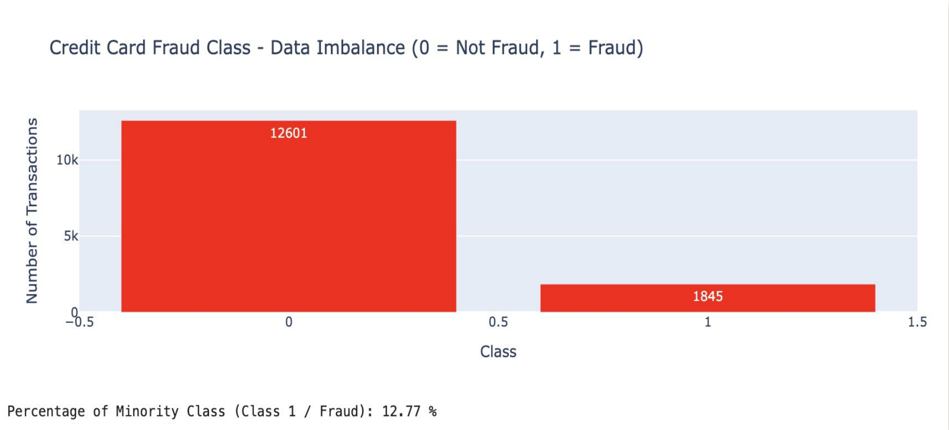
	merchant	category	city	state	job
14441	Hudson-Grady	shopping_pos	Athena	OR	Dealer
14442	"Nienow, Ankunding and Collie"	misc_pos	Gardiner	OR	"Engineer, maintenance"
14443	Pacocha-O'Reilly	grocery_pos	Alva	WY	"Administrator, local government"
14444	"Bins, Balistreri and Beatty"	shopping_pos	Wales	AK	"Administrator, education"
14445	Daugherty-Thompson	food_dining	Unionville	MO	"Investment banker, corporate"

ML models expects for **numerical input** and will ignore string values

Solution?
Apply the **variable encoding** to quantitative measures while preserve the qualitative meaning. Use the **Label Encoder**, useful for tree-based models

Data Imbalance Check

Count the number of **fraud** and **non-fraud** transaction



Why does it matter?

Any imbalances problem will have implication on both training and evaluation process. Data imbalance check aims to examine whether target variable classes have even distribution, given our data is binary classification.

83%

Data are dominated by **Class 0 or Non-Fraud transactions**

Biased towards majority class

Misleading accuracy metrics

Poor prediction on minority class

Solution ? Oversampling techniques

SMOTE

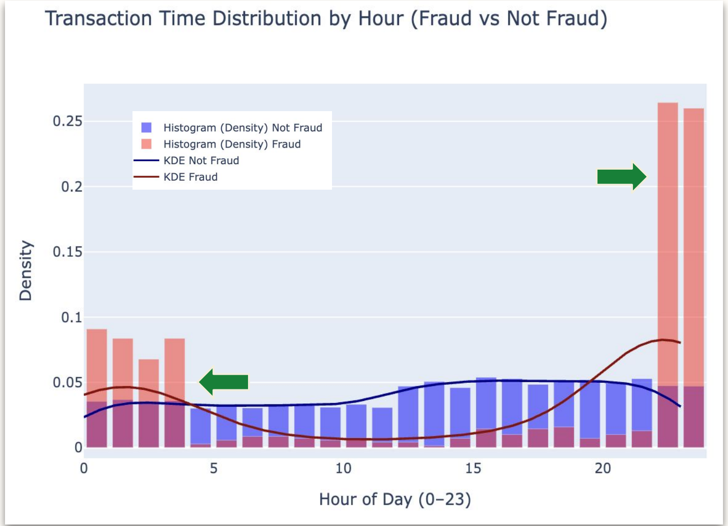
Increase sample of minority class by replicating existing examples of minor class but applied strictly **only for training dataset** to prevent **data leakage**

03

Exploratory Data Analysis (EDA)

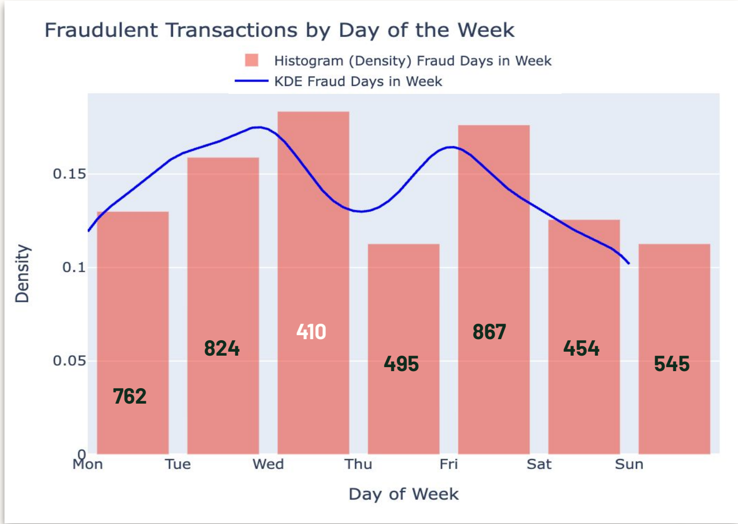
Data Distribution by Features

Transaction time in a day



- **Frauds** happened during **off-peak hours** when people are less alert → opportunities to attack
- Non-fraud transactions are evenly spread throughout the day

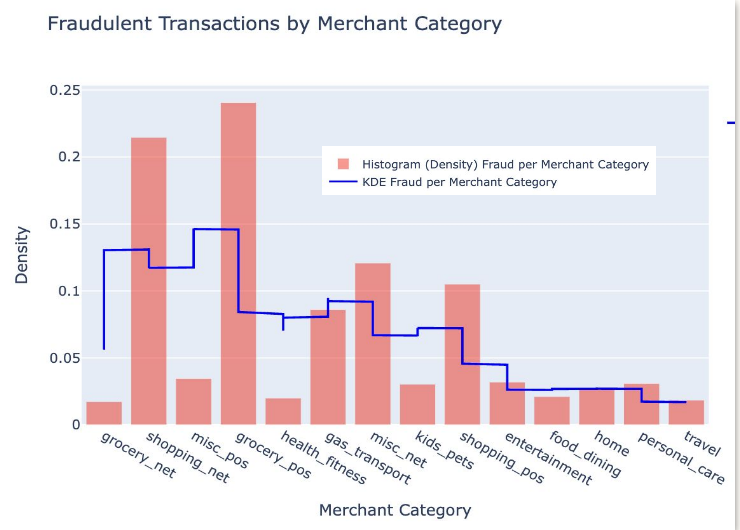
Weekend vs Weekday



- **Highest fraud** rate occurred **during weekdays** (e.g., Wed) despite having lowest transaction volume
- Indicates fraudsters **target low activity** period, possibly due to **merchants are less vigilant**

Data Distribution by Features

Merchant Category

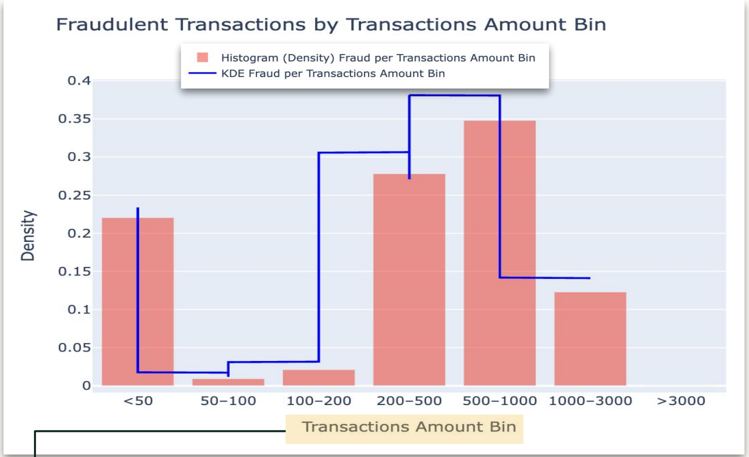


merchant_category	Total_Transactions_Per_Cat	Fraud_Transactions_Per_Cat	Fraud_Rate
grocery_pos	1602	444	28.0
shopping_net	1408	396	28.0
misc_net	821	223	27.0
shopping_pos	1354	194	14.0
gas_transport	1430	159	11.0
travel	385	34	9.0
misc_pos	823	64	8.0
grocery_net	474	32	7.0
entertainment	953	59	6.0
personal_care	990	57	6.0
kids_pets	1141	56	5.0
food_dining	870	39	4.0
health_fitness	891	37	4.0
home	1304	51	4.0

- **e-Commerce, offline groceries**, other online transactions has the highest fraud rate
- Fraudsters tend to target **online transactions** → does not have physical credit transaction system (e.g., EDC)
- **High frequency in grocery** transactions makes **frauds harder to be identified** → merchant prioritize on speed of purchasing process than verification process

Data Distribution by Features

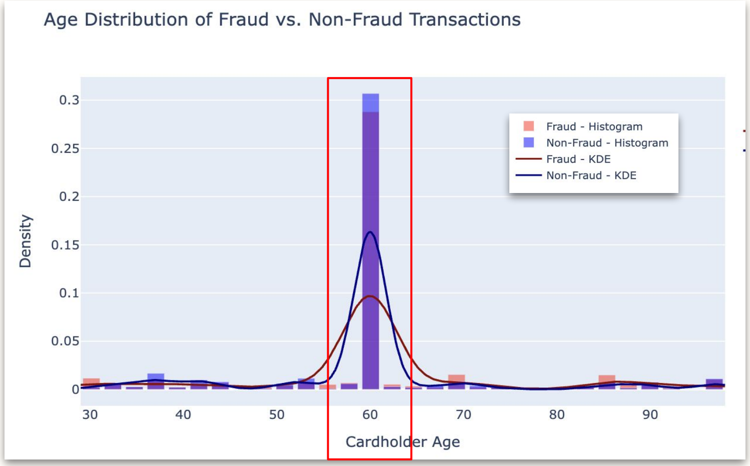
Transaction time in a day




```
# Step 1: Create new custom bins for transactions amount
transaction_bins = [0, 50, 100, 200, 500, 1000, 3000, np.inf]
transactions_labels = ['<50', '50–100', '100–200', '200–500',
                       '500–1000', '1000–3000', '>3000']
df_cleaned['amount_bin'] = pd.cut(df_cleaned['amt'],
                                  bins=transaction_bins,
                                  labels=transactions_labels, right=False)
category_bin_inverse = dict(enumerate(transactions_labels)) # to map bin index to label
```

- Transaction amount that falls in **'<50'**, **'200-500'** and **'500-100'** bin category have **most frequent fraud**

Weekend vs Weekday



Represents **68.8% of total** transactions → 9,957 transactions with **1,201 fraud transactions**

- Both fraud and non-fraud transactions are **concentrated** within credit cardholder **age of 60**
- Fraudsters are likely to target customer with **high vol transactions**

04

Model Building and Implementation

Using Supervised and Unsupervised Machine Learning Model

Data Preparation

For Supervised Learning

Data Splitting

X features:

- **Amount and location:** Amount, transactions bin, distance, city, state, city population
- **Time:** Hour, day of week, is weekend, missing time flag
- **Demographic:** Age, job
- **Merchant**: Merchant category

Y variable: Fraud binary label

!! Make sure all features are in numerical data type

Training 70%	Validation 15%	Test 15%

Split using **train_test_split** with stratify = Y to ensure the class proportions in the splitting data process are same for both training and test set

Data Resampling and Scaling

Step 1

Handle missing values in X variables to avoid error output in most ML models

```
imputer = SimpleImputer(strategy='median')
```

Fill with **median** from each column features

Imputer.fit_transform only for TRAINING SET

Imputer.transform for VALIDATION and TESTING SET

Step 2

Apply resampling method using SMOTE (for training set)

```
smote = SMOTE(random_state=42)
X_train_smote, Y_train_smote = smote.fit_resample(X_train_imputed, Y_train)
```

Step 3

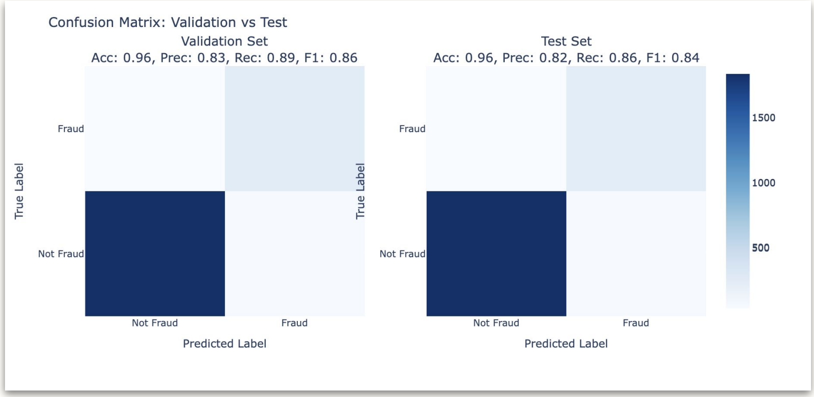
Do feature scaling to standardize features' scale

```
scaler = StandardScaler()
```

for **faster convergence** and **effective model learning**

Supervised Learning: Fraud Detection Model

Decision Tree



Model purpose:
Use for data classification prediction problems using ASM method to choose the best attributes (decision node) from historical pattern and repeat the process recursively until no more remaining attributes.

Advantages:

- Able to handle complex and non-linear relationships between transactional features and class labels
- Minimal preprocessing required and robust to outliers
- Creates clear "if-then" rules, hence interpretable

Performance Result and Analysis

Validation Set

```
Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.97      0.98      1890
           1       0.83      0.89      0.86       277

    accuracy                           0.96      2167
   macro avg       0.91      0.93      0.92      2167
weighted avg       0.96      0.96      0.96      2167

Accuracy: 0.9626211352099677
Precision: 0.8288590604026845
Recall: 0.8916967509025271
F1 Score: 0.8591304347826086
ROC AUC Score: 0.9323563119592
```

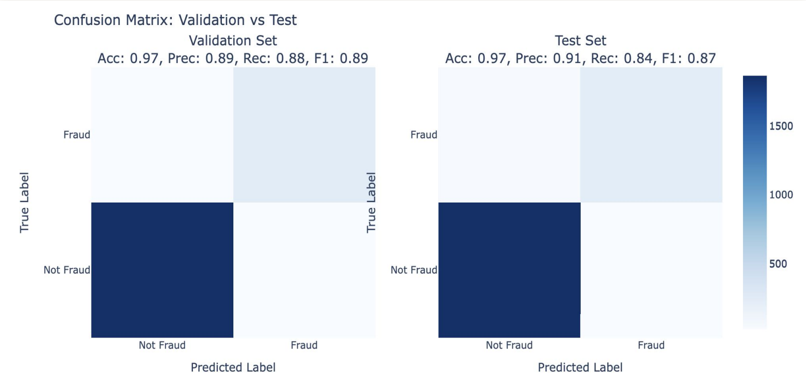
Testing Set

```
Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.97      0.98      1890
           1       0.82      0.86      0.84       277

    accuracy                           0.96      2167
   macro avg       0.90      0.92      0.91      2167
weighted avg       0.96      0.96      0.96      2167

Accuracy: 0.957544993077988
Precision: 0.8156996587030717
Recall: 0.8628158844765343
F1 Score: 0.8385964912280702
ROC AUC Score: 0.9171222279525528
```

- **Accuracy:** ~96% — correctly classifies most transactions.
- **Precision (Fraud):** ~83% — few false alarms; reduces disruption for real customers.
- **Recall (Fraud):** 86–89% — captures most fraud cases, minimizing financial loss.
- **ROC-AUC:** 0.91–0.93 — robust discrimination between fraud and legitimate activity.

Supervised Learning: Fraud Detection Model

Random Forest Classifier



Model purpose:
Uses ensemble method for generate classification predictions from multiple decision trees that created from different subsets of data and randomized features. It will take the majority voting from all trees as the final result, reduce the overfitting issues.

Advantages:

- Combine diverse decision trees ("forest") to capture complex, non-linear fraud patterns → higher accuracy
- Minimal preprocessing required and robust to outliers
- Able to work with high-multidimensional data features

Performance Result and Analysis

Validation Set

```
Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.98      0.98      1890
           1       0.89      0.88      0.89       277

    accuracy                           0.97      2167
   macro avg       0.94      0.93      0.94      2167
weighted avg       0.97      0.97      0.97      2167

Accuracy: 0.9713890170742963
Precision: 0.8937728937728938
Recall: 0.8808664259927798
F1 Score: 0.8872727272727273
ROC AUC Score: 0.9826208622237503
```

Testing Set

```
Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.99      0.98      1890
           1       0.91      0.84      0.87       277

    accuracy                           0.97      2167
   macro avg       0.94      0.91      0.93      2167
weighted avg       0.97      0.97      0.97      2167

Accuracy: 0.968158744808491
Precision: 0.90625
Recall: 0.8375451263537906
F1 Score: 0.8705440900562852
ROC AUC Score: 0.9831814795713713
```

- **Accuracy:** ~97% — correctly classifies majority transactions.
- **Precision (Fraud):** ~89% — minimizes false positives, reduce disruption for legitimate customers.
- **Recall (Fraud):** 84–88% — captures most fraud cases, minimizing financial loss.
- **ROC-AUC:** 0.98 — robust discrimination between fraud and legitimate activity.

Supervised Learning: Fraud Detection Model

XGBoost Classifier

Model purpose:

Builds models sequentially using gradient boosting, where each new tree corrects the errors made by previous weaker learners to minimize the overall loss function. It will combine all learned trees using weighted sum to generate final classification.

Advantages:

- Include built-in regularization (i.e., L1 and L2) hence less resistant to overfitting issues
- Fast and efficient for large scale datasets
- Able to handle missing values and imbalance data

Performance Result and Analysis

Validation Set

```
Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.99      0.99      1890
           1       0.93      0.90      0.91       277

    accuracy                           0.98      2167
   macro avg       0.96      0.94      0.95      2167
weighted avg       0.98      0.98      0.98      2167

Accuracy: 0.9778495616059067
Precision: 0.9288389513108615
Recall: 0.8953068592057761
F1 Score: 0.9117647058823529
ROC AUC Score: 0.9872566997115733
```

Testing Set

```
Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.99      0.99      1890
           1       0.93      0.86      0.89       277

    accuracy                           0.97      2167
   macro avg       0.96      0.93      0.94      2167
weighted avg       0.97      0.97      0.97      2167

Accuracy: 0.9741578218735579
Precision: 0.9333333333333333
Recall: 0.8592057761732852
F1 Score: 0.8947368421052632
ROC AUC Score: 0.9867877676541936
```

- **Accuracy:** ~97% — correctly classifies majority transactions.
- **Precision (Fraud):** ~93% — minimizes false positives, reduce disruption for legitimate customers.
- **Recall (Fraud):** 86–90% — captures most fraud cases, minimizing financial loss.
- **ROC-AUC:** 0.98 — robust discrimination between fraud and legitimate activity.

Data Preparation

For Unsupervised Learning

Variable Definition

X features:

- **Amount and location:** Amount, transactions bin, distance, city, state, city population
- **Time:** Hour, day of week, is weekend, missing time flag
- **Demographic:** Age, job
- **Merchant**: Merchant category

Y variable: Fraud binary label

!! Make sure all features are in numerical data type

Why not perform data splitting process?

No true labels to predict or test

Used for pattern discovery

Need to use **full dataset** to obtain the most information → **increase detection power**

Data Scaling

Step 1

Drop missing values in X variables

```
X_unsupervised = X_unsupervised.dropna()
Y_unsupervised = Y_unsupervised.loc[X_unsupervised.index]
```

To preserve the **true structure of data** and **avoid bias** introduced by imputation method due to absence of true label

---

Step 2

Do feature scaling to standardize features' scale

```
scaler = StandardScaler()
X_unsuper_scaled = scaler.fit_transform(X_unsupervised)
```

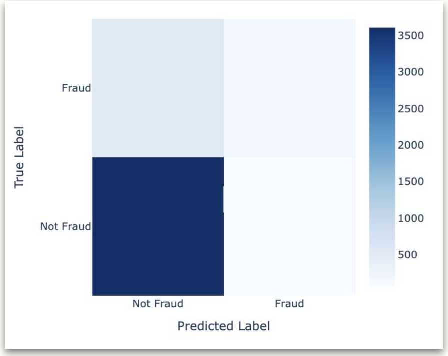
for **faster convergence** and **effective model learning**

No SMOTE resampling applied in unsupervised learning process

Unsupervised Learning: Anomaly Detection Model

Isolation Forest

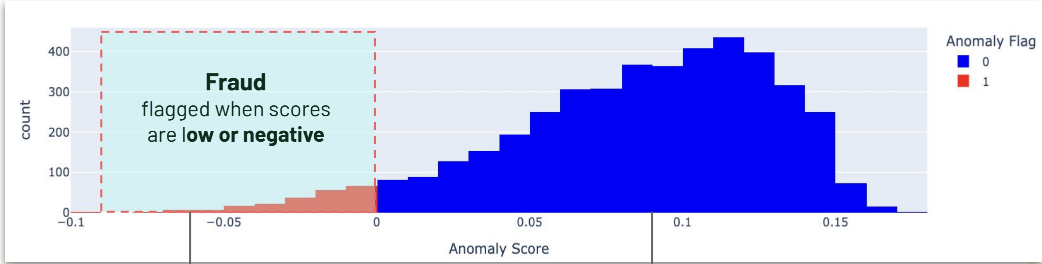
Confusion Matrix



False Negative (FN)
531 fraud transactions are missed

True Positives (TP)
161 transactions correctly identified as fraud

Anomaly Score Distribution



.decision_function() will return anomaly scores with lower values = more anomalous

More distribution clustered on the **right side**, indicating majority class is dominating and **anomalies are rare** (i.e., seen as different from rest)

Model purpose:
Detects anomalies by isolating data points through randomly generated decision trees, where anomalies (i.e., ones with unusual feature values) are easier to isolate due to fewer splits required.

Advantages:

- Not relying on the labeled data, hence suitable for limited fraud labels in real world scenario
- Able to handle data with multiple features or anomalies
- Linear time complexity and unsupervised

```
=== Isolation Forest Model Performance ===
Classification Report:
              precision    recall  f1-score   support

           0       0.87      0.98      0.92      3665
           1       0.74      0.23      0.35       692

    accuracy                           0.87      4357
   macro avg       0.81      0.61      0.64      4357
weighted avg       0.85      0.87      0.83      4357

Accuracy: 0.8650
Precision: 0.8506
Recall: 0.8650
F1 Score: 0.8340
ROC AUC Score: 0.8261
```

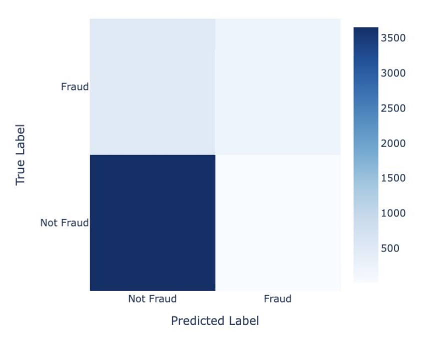
Model has **good precision** on frauds detection, but **low recall** (23%). This meaning that most frauds are missed due to lack of labels during training.

Treated as for early-stage fraud filtering.

Unsupervised Learning: Anomaly Detection Model

Auto Encoders

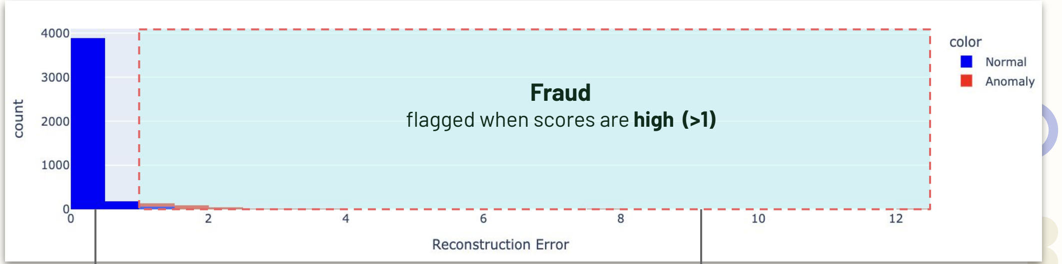
Confusion Matrix



False Negative (FN)
489 fraud transactions are missed

True Positives (TP)
203 transactions correctly identified as fraud

Reconstruction Error Distribution



Distribution more clustered to **left side** as majority class will have **low error values**

Not much showing flagged anomaly distribution, indicating **low recall issues**

Model purpose:
Learns compressed representations of normal transactions pattern through encoding and decoding, and flags frauds based on high reconstruction errors (i.e., those with high discrepancy to normality)

Advantages:

- Able to handle complex and non-linear relationships pattern
- Effective for high-dimensional data with many features involved
- Clear separation rules → normal vs non-normal through error values

```
=== Auto Encoder Model Performance ===
Classification Report:
              precision    recall  f1-score   support

           0       0.88      1.00      0.94      3665
           1       0.93      0.29      0.45       692

    accuracy                           0.88      4357
   macro avg       0.91      0.64      0.69      4357
weighted avg       0.89      0.88      0.86      4357

Accuracy: 0.8843
Precision: 0.8897
Recall: 0.8843
F1 Score: 0.8577
ROC AUC Score: 0.8517
```

Model has **strong precision** on frauds detection, but **low recall** (29%). This meaning that most frauds are missed due to lack of labels during training.

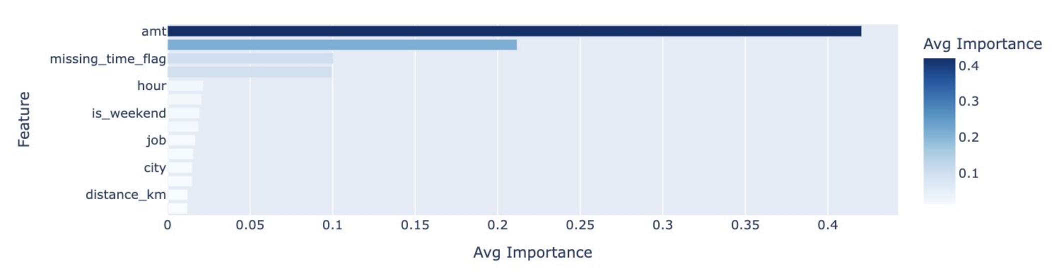
Could be used for high-precision filter in multi-stage detection phase

Feature Importance Analysis

Supervised Learning Model

Based on Decision Tree and Random Forest

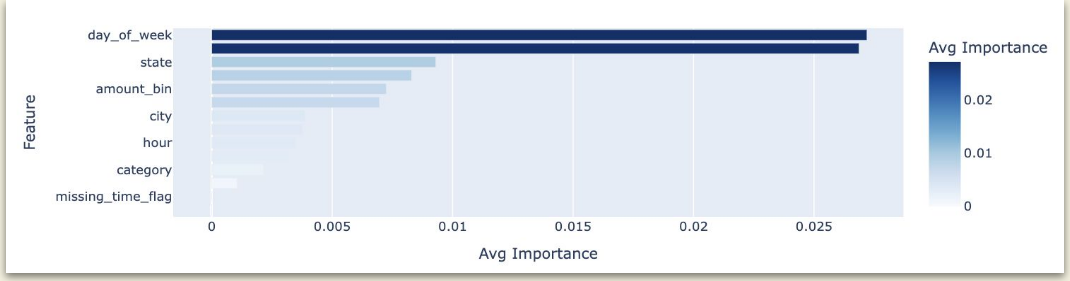
- **Transaction amount (i.e., 'amt' and 'amt_bin')** as the dominant predictor of fraud,
- **Timing-related features** (**i.e., missing_time_flag** and **hour**) also become second contributor



Unsupervised Learning Model

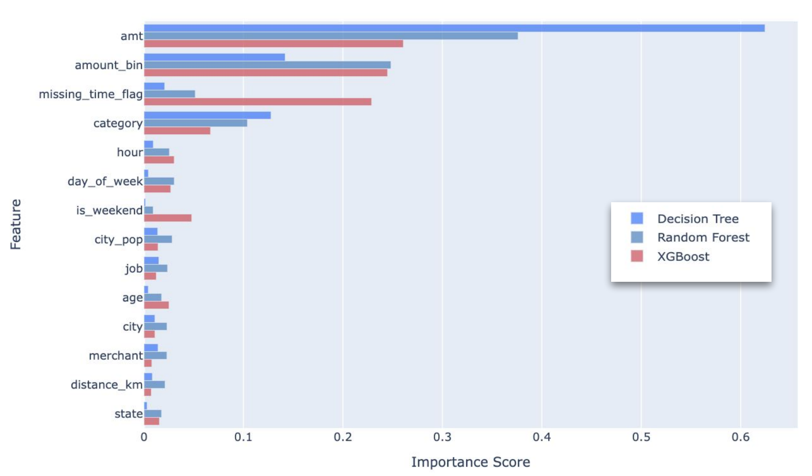
Based on Isolation Forest and AutoEncoder

- **Days and location (i.e., 'state', 'city')** suggest key signals for detecting anomalies pattern
- **Transaction amount ('amount_bin')** also become second importance



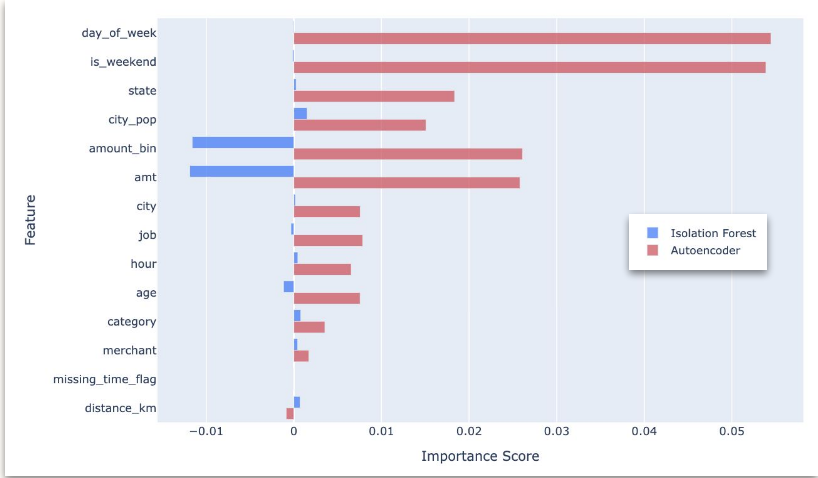
Feature Importance Analysis

Supervised Learning Model



- **Transaction amount** is the most influential feature for all models
- **Ensemble methods** (i.e., Random Forest and XGBoost) assign higher absolute importance scores since they combine multiple trees to generate more robust and complex use of features
- **Missing time flag** is become second strong indicator for XGBoost model predictor

Unsupervised Learning Model



- **Transaction amount** is consistently identified as high importance features for both model despite different direction sign
- **Negative scores** generated by Isolation Forest meaning those features might decrease the ability to identify anomalies
- **Autoencoders** is more sensitive for detecting **temporal/contextual** fraud patterns, while **Isolation Forest** focuses on **financial patterns**.
- Combining both improves anomaly coverage by capturing diverse fraud types.

05

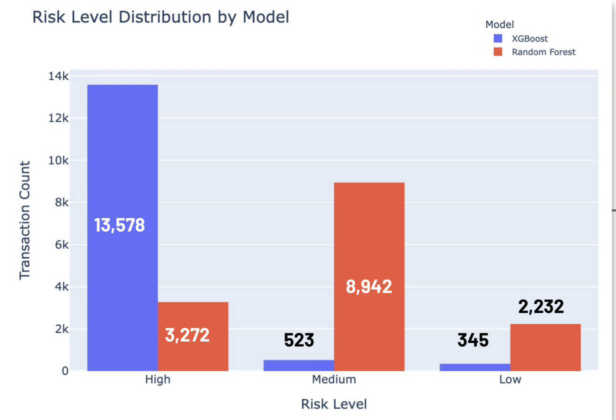
Risk Scoring Analysis

Fraud Risk Scoring

Model Based Method

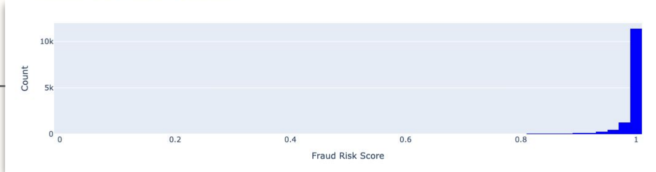
What does the model do? It will quantify the **probability** of transactions being fraudulent, in other words the **risk of fraud** transactions. This is performed using **trained** supervised learning model and apply the **predict_proba( )**.

What is the risk category ? We assign **high risk** for scores > 0.9, **medium risk** for 0.6-0.9, and **low risk** for <0.6 (*subject to user judgment*)



Contradicting results from each model are due to different model sensitivity to heavy class imbalance data - **conservative vs extreme** assignment

XGBoost



Assigns almost all transactions a very high fraud risk score, suggesting it is extremely **aggressive**

Random Forest

Assigned scores are more spreaded out, concentrating to medium to high risk – more **conservative** assessment

There are differences in how model handle the fraud **probability calibration** or **splitting criteria**

Fraud Risk Scoring

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Misalignment Result

1 **Issue 1:** High vs low score

	is_fraud	xgb_fraud_risk_score	rf_fraud_risk_score
0	1	0.999802	0.90
1	1	0.919799	0.52
2	1	0.999936	0.92
3	1	0.999957	0.93
4	1	0.999993	0.91

2 **Issue 2:** High for non-fraud label

	is_fraud	xgb_fraud_risk_score	rf_fraud_risk_score
14441	0	0.762603	0.30
14442	0	0.998123	0.76
14443	0	0.998979	0.73
14444	0	0.047159	0.50
14445	0	0.995223	0.73

What is the underlying cause?

Too much **false positives**...here

	xgb_fraud_risk_score	is_fraud
1782	0.972581	0
1783	0.998009	0

	rf_fraud_risk_score	is_fraud
2019	0.97	0
2028	0.91	0

11,987 and 2,660 legitimate transactions are flagged with high risk by XGB and RF model respectively

Suggesting that...

- **Severe class imbalance -** models are more prone to **overfit** the non-fraud / majority class
- **Lack of probability calibration -** Tree-based models may output **overconfident** scores, making their predictions less reliable.

How to solve?

Data Resampling

Model is able to learn equal class distribution, hence **stabilize the probability** output for each model and allow for more **comparability** and **consistent** risk scores

Fraud Risk Scoring

Model Based Method

Validation Set Performance

When RF model scores > 0.9 but fraud label = 0

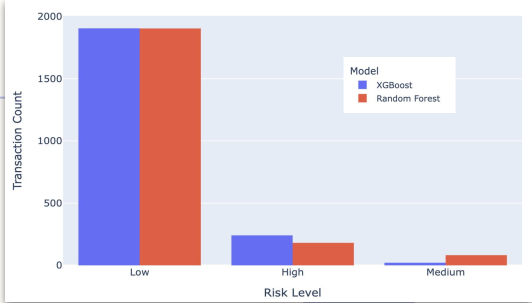
	rf_fraud_risk_score	xgb_fraud_risk_score	is_fraud
422	0.91	0.908079	0
471	0.92	0.961016	0

Number of False Positives (FP) under RF = 8

When XGB model scores > 0.9 but fraud label = 0

	rf_fraud_risk_score	xgb_fraud_risk_score	is_fraud
1316	0.70	0.949410	0
1411	0.75	0.986665	0

Number of False Positives (FP) under XGB = 10



More consistent results with small False Positives numbers

Data	RF	XGB
Validation	H = 181 M = 82 L = 1,904	H = 241 M = 21 L = 1,905
Test	H = 178 M = 70 L = 1,919	H = 234 M = 17 L = 1,916

Testing Set Performance

When RF model scores > 0.9 but fraud label = 0

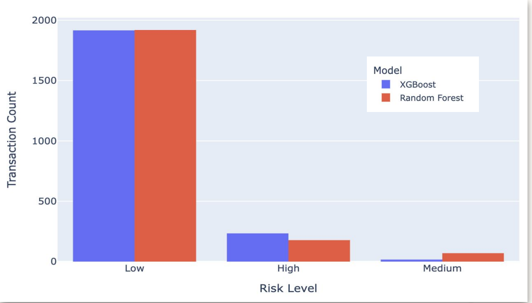
	rf_fraud_risk_score	xgb_fraud_risk_score	is_fraud
678	0.99	0.999416	0
838	0.97	0.992667	0

Number of False Positives (FP) under RF = 8

When XGB model scores > 0.9 but fraud label = 0

	rf_fraud_risk_score	xgb_fraud_risk_score	is_fraud
121	0.85	0.987875	0
557	0.44	0.954852	0

Number of False Positives (FP) under XGB = 10



Fraud Risk Scoring

Rule Based Method

What does the model do? It will assign the **probability** based on the pre-defined logical **("if-then") rules** using **risk factors** from the EDA analysis step. Risk factors used are from the important features and with each rule checks for any pattern beyond threshold.

What is the risk category ? Our strategy is to assign **up to 10 points**, with each risk factors have their own contribution (e.g., **high risk = high points**)

Applied Scoring System

Features	Risk Level	Scoring Point	Rationale
Amount (Key Factor)	High	4-6 points for large transactions	53% fraud rate in 500~1000 bin, 88% fraud >1000 bin
Merchant Category (Key Factor)	High	3 points in high-risk category	shopping_net, misc_net show highest fraud density
Transaction Time	Medium	2-3 points for late/early hours	0~3 AM peak fraud density, followed by 22~23 PM
Distance	Low	1 point	Abnormal distances (>100km) may indicate fraud
Missing Time Stamp	Low	1 point	Missing transaction time data linked to anomalies
Credit Holder Age	Low	1 point	High transaction volume at age 60~65

Fraud Risk Scoring

Rule Based Method

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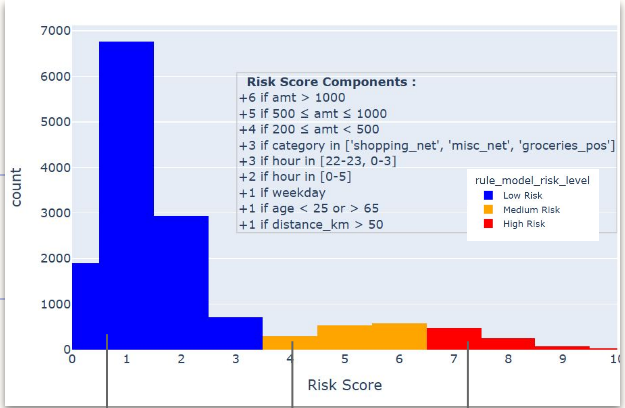
Risk Score Distribution

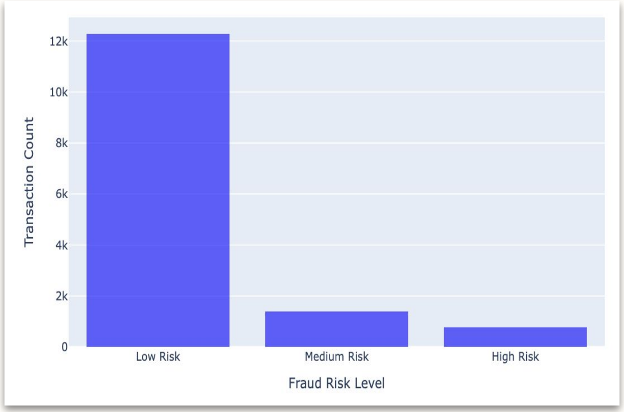


Likely **normal** transactions with **regular** behavior

Slightly **unusual** behavior with trigger on **few rules**

Likely to be **fraud** that trigger **multiple rules**

Risk Level Distribution



Based on risk score distribution...

Most transactions are assigned with low risk probability, suggesting normal / legitimate. While, high scorers may reveal layered risk signals trigger, suitable for further investigation or fraud alerts.

Based on risk level distribution...

Aggregating the distribution, this agree with our dataset that most of transactions are labeled as non-fraud.

*The rule-based model prioritizes **minimizing false alarms** using clear rules, offering **more transparent** detection system and **incorporate domain knowledg**e as the rule set basis...*

Fraud Risk Scoring

Hybrid (Rule-Enhanced) Risk Score Model

What does the model do? Combine **Random Forest** predictions with **domain expert rules** to adjust risk scores for high-risk scenarios, improving interpretability and robustness of fraud detection as post-calibration techniques.

What is the risk category ? Our strategy is to adjust the base score and apply **risk-enhancing rule** conditions with total score adjustment capped to 1

Applied Scoring System

Features	Risk Level	Scoring Point Adjustment	Rationale
Amount	High	+0.2 points	For amount > 1,000
Merchant Category	High	+0.2 points	For online shopping, offline groceries
Transaction Time	Medium	+0.1 points	For time 12am-3am and 10-11pm
Distance	Low	+0.05 points	For above 100km
Missing Time Stamp	Low	+0.1 points	For transactions without time stamp info
Credit Holder Age	Low	+0.05 points	For age below 25 or above 65

Fraud Risk Scoring

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Risk Score Distribution

The **Hybrid Model** is more confident in assign high risk scores, **clearly separating** risk and safe transactions. Meanwhile **Random Forest** more spreaded out with transactions are **overlapping into medium** scores

Risk Level Distribution

More transactions are classified as **high risk,** and significant **lower portion** on **low risk** levels

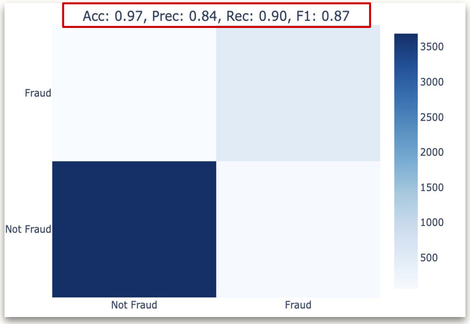
Key takeaways?

- **Hybrid model** is more **sensitive to risk signals**, capturing broader range of potentially risky transactions due to fixed set rules → improve detection precision
- The model will produce **clear signal,** help users to implement strict fraud boundaries while keep **minimizing the false positive** alarms
- Set rules could be **customized** and **scalable** according to users type.

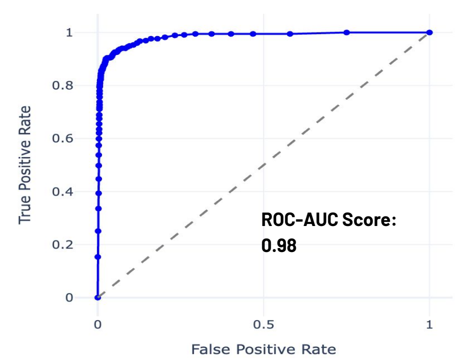
Risk Score Model Performance Evaluation

Random Forest Model

Confusion Matrix

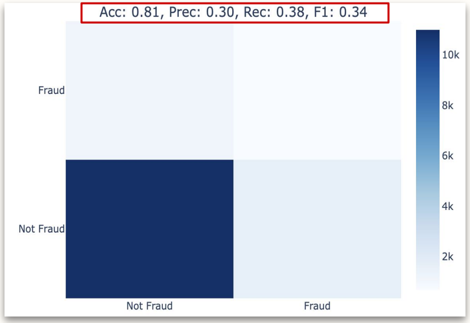


ROC Curve

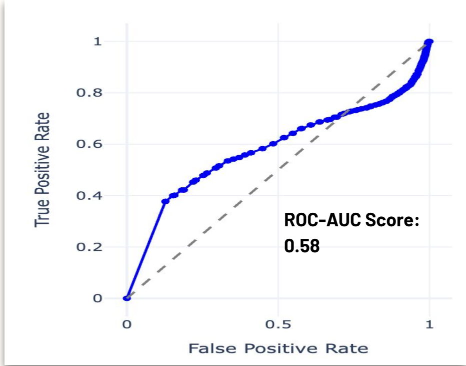


Hybrid Model

Confusion Matrix



ROC Curve



Top 10% High Risk Transaction Analysis

Among them, how many **actual frauds** are identified?

Random Forest	Rule-Based	Hybrid Model
19.58% transactions	**82.69%** transactions	**30.26%** transactions

Hybrid model improves RF results (i.e., **more true frauds**) but **less extreme** than rule-based approach…

Key Takeaways

- **Explainability Gap in Supervised Model (RF)**
  Able to detect some frauds but miss subtle patterns and lack in interpretability (i.e., black box)
- **High Accuracy but Low Flexibility in Rule-Based**
  It could capture frauds effectively but manual rules may need further review to become scalable and flexible
- **Limited Enhancement in Hybrid Model**
  Better than RF alone but not stronger than rule-based, any conflicts between supervised and rule would limit its predictive ability

Thank You

Any questions?